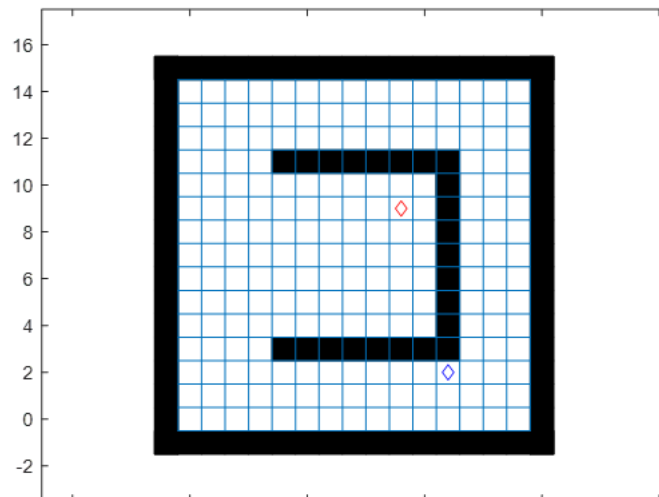
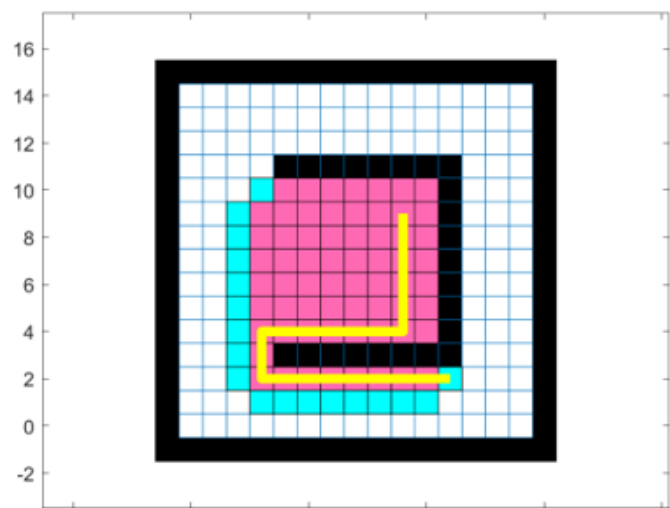


A-star algorithm can be used for the solution of many problems as well as optimization efficiently due to it is a heuristic search, which can improve the searching efficiency by decrease the nodes that need to be checked using heuristic function.

A algorithm is a method to find the shortest path. The goal of the algorithm is to find the shortest path.



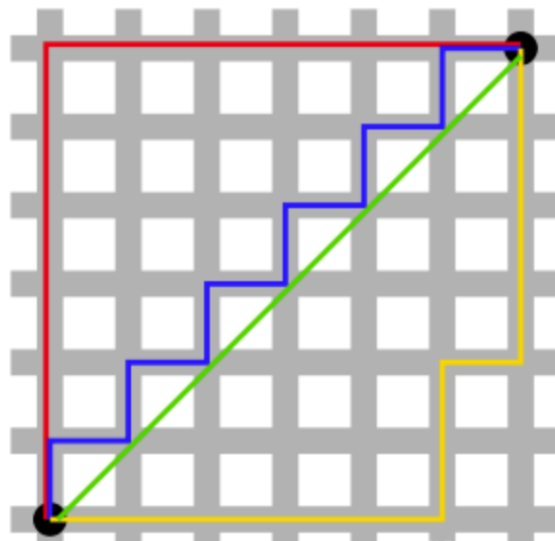
The A-star algorithm searches the shortest path. The center point of the grid is used to represent the positions of the nodes. The openList is to store the elements which will be



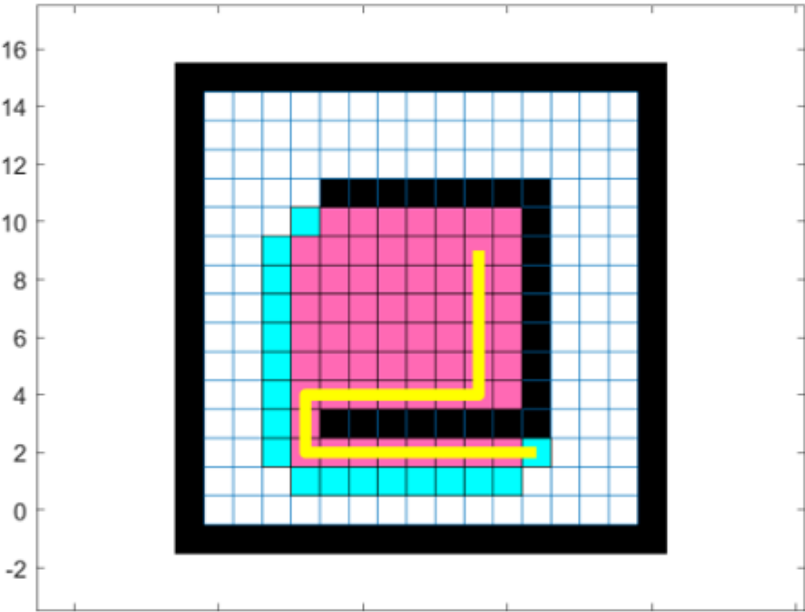
MCEE 2023 Volume 46 (2023) 157 2.2. Basic model 2.2.1. Raster method 2.2.2.
Manhattan distance formula Manhattan distance is used to calculate the value of the

nodes in geometric metric space. The value of Manhattan distance in the 2-dimensional plane is the value between two points, calculating from the vertical axis and the distance on the horizontal axis and adding these two distances. 2.2.2. Manhattan distance formula
Manhattan distance is used to calculate the value of the absolute axis distances of two

The green line is the Euclidean distance, which is used to calculate the straight-line distance



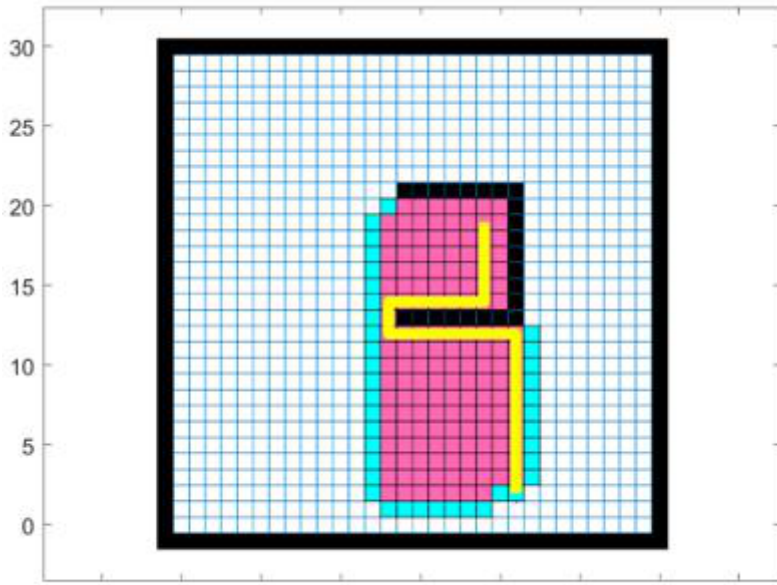
The goal of this study is to find the shortest path.



Find the starting node. Find the current node. Find the path.

This paper uses the Raster Method to implement the environment and set the obstacle and starting point and target automatically. The algorithm uses the evaluation function to

the future and selects the points to be prioritized for walking, which works well in searching the optimized shortest path.



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